

AIMO Proof Pilot — Grading Sheet

Problem (#_ID): *[e.g. 4_COLOUR]*

Submission: *[e.g. team_3 / model_2]*

Grader initials: *[e.g. SB]*

Date: *[YYYY-MM-DD]*

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (`docs/markschemes.pdf`).

1. Proposed mark

Proposed mark (choose one): ☐0 ☐1 ☐6 ☐7

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

[Your reasoning here.]

Significantly different approach? If the solution is significantly different from the reference solution, map it onto the markscheme milestones as best you can, state your rationale here, and flag it for discussion.

[If applicable: how you matched the approach to the milestones, and your rationale. Otherwise write “N/A”.]

3. Errors / issues

List **every** error or issue you find, *even minor ones*, with the corresponding line number(s) in the submission. The *short summary* should be a brief phrase (we will later turn these into reusable codes); the *detailed description* should give the full explanation. Mark each as:

- *Trivial* – those listed under NME which we won't penalise;
- *Minor* – those listed under ME which could turn $7 \rightarrow 6$; or
- *Major* – those listed under NEC which would drop a solution to 0/1

guided by the markscheme.

Line(s)	Short summary	Detailed description	Severity
<i>[e.g. 12–15]</i>	<i>[brief phrase]</i>	<i>[full explanation]</i>	<i>[trivial / minor / major]</i>

(Add rows as needed.)

4. Proof quality (qualitative)

These two questions capture *stylistic* qualities of the write-up for our later analysis. They do **not** affect the 0/1/6/7 mark — answer them independently of the score. Choose one option per question (the examples are there to keep graders consistent).

4a. Readability

How much difficulty did you have following the proof? (Was it easy to read, or did you have to re-read passages to understand them?)

Choose one: ☐ None ☐ A bit ☐ A lot

- **None** — read smoothly in a single pass; steps, notation, and structure were clear throughout.
- **A bit** — mostly clear, but a few passages needed re-reading (e.g. a terse step, ambiguous notation, or a slightly confusing ordering).
- **A lot** — frequently hard to follow; had to re-read many parts, reconstruct missing links, or untangle a disorganised argument.

[Optional: a sentence or two on the worst readability issues, with line numbers.]

4b. Conciseness

How much unnecessary or inefficient content was there? (Were lemmas proved efficiently, or was there significant irrelevant / repetitive material?) Judge **local** optimisations only — e.g. a sub-argument proved more verbosely than needed, or repetition — *not* whether the overall approach takes a globally sub-optimal route.

Choose one: ☐ None ☐ A bit ☐ A lot

- **None** — tight throughout; lemmas proved efficiently, with little or no wasted content.
- **A bit** — some redundancy or slightly laboured steps, but nothing major (e.g. a lemma proved more verbosely than necessary, or a little repetition).
- **A lot** — substantial irrelevant or repetitive content, or several sub-arguments proved far less efficiently than they could have been.

[Optional: a sentence or two on the main conciseness issues, with line numbers.]